

Group Name:	AbsTrack Fitness Gym: Gym Management System
Proponents:	Rosalada Justin Troy Merka Nathaniel Keene Felecilda Lloyd Justin Sexcion Eulo Icon
Module Name:	Membership Management System

CL17L: SOFTWARE ENGINEERING 2

Milestone 6: Functional test case specification

SUMMARY OF TEST CASES EXECUTED

SCENARIOS	EXPECTED RESULT	ACTUAL RESULT	TEST RESULT (Passed/F ailed)	ACTION TAKEN
Create Member: Can't create member if required field/s are empty.	Display warning: "Please check the form for errors." and highlight empty field/s.	Displayed warning and highlighted empty field/s.	Passed	
Create Member: Can't create member if name field is empty.	Display validation error: "The name field is required."	Validation error displayed for name field.	Passed	
Create Member: Can't create member if contact field is empty.	Display validation error: "The contact field is required."	Validation error displayed for contact field.	Passed	
Create Member: Can't create member if contact format is invalid (contains letters).	Display validation error: Contact must match phone number format.	Validation error displayed correctly.	Passed	
Create Member: Can't create member if plan type is not selected.	Display validation error: "The plan type field is required."	Warning displayed correctly.	Passed	

Create Member: Can't create member if start date is empty.	Display validation error: "The start date field is required."	Validation error displayed correctly.	Passed	
Create Member: Can't create member if due date is empty.	Display validation error: "The due date field is required."	Validation error displayed correctly.	Passed	
Create Member: Can't create member if due date is before start date.	Display validation error: "The due date must be a date after start date."	Validation error displayed correctly.	Passed	
Create Member: Can't create member if exact same name already exists.	Display warning: "A member with the exact name '[name]' already exists."	Displayed duplicate name warning.	Passed	
Create Member: Warning shown if similar name (same first name) already exists.	Display warning: "A member with a similar name '[name]' already exists. Please confirm this is a different person."	Displayed similar name warning with confirmation prompt.	Passed	
Create Member: Create member if similar name confirmed and all inputs are valid.	Display: "Membership created successfully!" and return to Membership page.	Member saved and success message displayed.	Passed	
Create Member: Create member if OK button is clicked and all inputs are valid.	Display: "Membership created successfully!" and return to Membership page.	Member saved and success message displayed.	Passed	
Create Member: Can't create member if avatar file exceeds 2MB.	Display validation error: "Avatar must not exceed 2MB."	Validation error displayed correctly.	Passed	
Create Member: Can't create member if avatar file type is not jpeg/jpg/png/gif.	Display validation error: "Avatar must be an image of type jpeg, jpg, png, or gif."	Validation error displayed correctly.	Passed	

Update Member: Can't update member if required field/s are cleared and left empty.	Display warning: "Please check the form for errors." and highlight empty field/s.	Warning displayed and empty fields highlighted.	Passed	
Update Member: Update member if valid changes are made and UPDATE is clicked.	Display: "Membership updated successfully!"	Member details updated successfully.	Passed	
Update Member: Status auto-recalculates to "Active" when the due date is updated to the far future.	Member status changes to "Active."	Status updated correctly.	Passed	
Update Member: Status auto-recalculates to "Due soon" when the due date is updated to within 7 days.	Member status changes to "Due soon."	Status updated correctly.	Passed	
Update Member: Status auto-recalculates to "Expired" when the due date is updated to a past date.	Member status changes to "Expired."	Status updated correctly.	Passed	
Delete Member: Can't delete member if CANCEL button is clicked.	No changes happen.	No changes occurred.	Passed	
Delete Member: Delete member if OK/confirm button is clicked.	Display: "Membership deleted successfully!"	Member deleted successfully.	Passed	
Delete Member: Can't delete a member that doesn't exist (ID not found).	Display: "Membership not found."	Error message displayed correctly.	Passed	
Bulk Delete: Can't bulk delete if no members are selected.	Display: "No memberships selected for deletion."	Warning displayed correctly.	Passed	

Bulk Delete: Bulk delete selected members if confirmed.	Display: "Successfully deleted [n] membership(s)."	Members deleted and count shown.	Passed	
Renew Member: Can't renew if due date is before or equal to start date.	Display error: "Due date must be after start date."	Error displayed correctly.	Passed	
Renew Member: Renew membership with valid dates.	Display: "Membership renewed successfully! New due date: [date]."	Membership renewed and new due date shown.	Passed	
Search Member: Search by name returns matching results.	Matching members displayed in the list.	Correct members displayed.	Passed	
Search Member: Search with no matching results.	Empty list displayed, no members shown.	No results shown correctly.	Passed	
Membership Expiry: Automatically set status to Expired if due date has passed.	Member status changes to "Expired."	Status updated correctly.	Passed	
Membership Due Soon: Automatically set status to "Due soon" if due date is within 7 days.	Member status changes to "Due soon."	Status updated correctly.	Passed	

Group Name:	AbsTrack Fitness Gym: Gym Management System
Proponents:	Rosalada Justin Troy Merka Nathaniel Keene 89Felecilda Lloyd Justin Sexcion Eulo Icon
Module Name:	Attendance Management System

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SUMMARY OF TEST CASES EXECUTED

SCENARIOS	EXPECTED RESULT	ACTUAL RESULT	TEST RESULT (Passed/Failed)	ACTION TAKEN
Check-In: Can't check in if no client is selected.	Display validation error: "Please select a client."	Validation error displayed correctly.	Passed	
Check-In: Can't check in if client has already checked in today.	Display warning: "This client has already checked in today."	Duplicate check-in warning displayed.	Passed	
Check-In: Check in successfully if all fields are valid and client hasn't checked in today.	Display: "Attendance recorded successfully!"	Attendance recorded and success message shown.	Passed	
Delete Attendance: Delete attendance record if confirmed.	Display: "Attendance record deleted successfully!"	Attendance record deleted successfully.	Passed	
Delete Attendance: Can't delete attendance if record doesn't exist.	Display error: "Failed to delete attendance."	Error message displayed.	Passed	
Bulk Delete Attendance: Can't bulk	Display error: "No records selected."	Warning displayed correctly.	Passed	

delete if no records are selected.				
Bulk Delete Attendance: Bulk delete selected attendance records if confirmed.	Display: "[n] attendance record(s) deleted successfully."	Records deleted and count shown.	Passed	
Create PT Schedule: Can't create if no client is selected.	Display validation error: "Please select a client."	Validation error displayed correctly.	Passed	
Create PT Schedule: Can't create if no trainer is selected.	Display validation error: "Please select a trainer."	Validation error displayed correctly.	Passed	
Create PT Schedule: Can't create if scheduled date is empty.	Display validation error: "Please select a date."	Validation error displayed correctly.	Passed	
Create PT Schedule: Can't create if scheduled date is in the past.	Display validation error: "Date cannot be in the past."	Validation error displayed correctly.	Passed	
Create PT Schedule: Can't create if scheduled time is empty.	Display validation error: "Please select a time."	Validation error displayed correctly.	Passed	
Create PT Schedule: Can't create if payment type is not selected.	Display validation error: "Please select a payment type."	Validation error displayed correctly.	Passed	
Create PT Schedule: Create successfully if all inputs are valid.	Display: "PT Schedule created successfully!"	PT schedule saved and success message shown.	Passed	
Update PT Schedule: Update PT schedule with valid changes.	Display: "PT Schedule updated successfully!"	PT schedule updated successfully.	Passed	
Update PT Schedule: Can't update with invalid payment type.	Display validation error for payment type.	Validation error displayed correctly.	Passed	
Update PT Status: Mark PT	Display: "Status updated to Done!"	Status changed to done.	Passed	

schedule as "done".				
Update PT Status: Mark PT schedule as "cancelled".	Display: "Status updated to Cancelled!"	Status changed to cancelled.	Passed	
Update PT Status: Can't update to invalid status value.	Display error: "Invalid status."	Error displayed correctly.	Passed	
Delete PT Schedule: Delete PT schedule if confirmed.	Display: "PT Schedule deleted successfully!"	PT schedule deleted successfully.	Passed	
Bulk Delete PT: Can't bulk delete if no schedules are selected.	Display error: "No records selected."	Warning displayed correctly.	Passed	
Bulk Delete PT: Bulk delete selected PT schedules if confirmed.	Display: "[n] PT schedule(s) deleted successfully."	Schedules deleted and count shown.	Passed	
Book Next Session: Book next session using last session's trainer and payment info.	Display: "Next session booked successfully!" with the previous trainer pre-filled.	Session booked with previous trainer info.	Passed	
Book Next Session: Book next session when client has no previous sessions (defaults).	Display: "Next session booked successfully!" with the trainer as "TBA" and payment as "Cash".	Session booked with default values.	Passed	
Search Attendance: Search attendance by client name.	Matching attendance records displayed.	Correct records displayed.	Passed	
Filter Attendance: Filter attendance by status.	Only attendance records matching status displayed.	Filtered correctly.	Passed	
KPI Dashboard: Display correct customer entered today count.	KPI shows accurate count of today's check-ins.	Correct count displayed.	Passed	

Group Name:	AbsTrack Fitness Gym: Gym Management System
Proponents:	Rosalada Justin Troy Merka Nathaniel Keene Felecilda Lloyd Justin Sexcion Eulo Icon
Module Name:	Payment and Billing Management System

CL17L: SOFTWARE ENGINEERING 2
Milestone 6: Functional test case specification

SUMMARY OF TEST CASES EXECUTED

SCENARIOS	EXPECTED RESULT	ACTUAL RESULT	TEST RESULT (Passed/Failed)	ACTION TAKEN
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Create Payment: Can't create payment if customer name is empty.	Display validation error: "The customer name field is required."	Validation error displayed correctly.	Passed	
Create Payment: Can't create payment if transaction type is empty.	Display validation error: "The transaction type field is required."	Validation error displayed correctly.	Passed	
Create Payment: Can't create payment if payment method is empty.	Display validation error: "The payment method field is required."	Validation error displayed correctly.	Passed	
Create Payment: Can't create payment if paid amount is empty.	Display validation error: "The paid amount field is required."	Validation error displayed correctly.	Passed	
Create Payment: Can't create payment if items data is empty (no items added).	Display validation error: "The items data field is required."	Validation error displayed correctly.	Passed	
Create Payment: Can't create payment if paid amount is less than total amount.	Display error: "Paid amount must be equal to or greater than total amount."	Error displayed correctly.	Passed	
Create Payment: Can't create payment if inventory has insufficient stock.	Display error: "Insufficient stock for [product name]."	Insufficient stock error displayed.	Passed	
Create Payment: Create payment successfully if all inputs valid and stock sufficient.	Display: "Payment completed successfully." and return to Payments page.	Payment recorded and success message displayed.	Passed	
Create Payment: Correct change/return amount calculated (paid - total).	Return amount shows correct calculation.	Correct return amount shown.	Passed	
Create Payment: Inventory stock decremented after successful	Product stock quantity reduced	Stock decremented correctly.	Passed	

payment.	by purchased amount.			
Create Payment: Receipt number auto-generated sequentially.	Receipt number auto-assigned (e.g., 0001, 0002, etc.).	Receipt number generated correctly.	Passed	
View Receipt: View receipt details for an existing payment.	Receipt displays customer name, items, amounts, and date.	Receipt details displayed correctly.	Passed	
View Receipt: Can't view receipt for a non-existent payment.	Display error: Payment not found.	Error displayed correctly.	Passed	
Delete Payment: Can't delete payment that doesn't exist.	Display error: "Failed to delete transaction."	Error displayed correctly.	Passed	
Bulk Delete: Bulk delete selected transactions and restore stock.	Display: "[n] transaction(s) deleted successfully." and inventory restored.	Transactions deleted and stock restored.	Passed	
Refund Payment: Refund a payment successfully with a reason.	Display success message and refund recorded.	Refund processed successfully.	Passed	
Refund Payment: Refund a payment with no reason provided (reason is optional).	Display success message, refund recorded with null reason.	Refund processed successfully.	Passed	
Refund Payment: Can't refund a payment that has already been refunded.	Display error: Payment already refunded.	Error displayed correctly.	Passed	
Payment History: View product payment history.	Product payments list displayed with details.	History displayed correctly.	Passed	
Payment History: View	Membership payments list	History displayed correctly.	Passed	

membership payment history.	displayed with details.			
Payment History: View combined refunded transactions.	Refunded transactions from both product and membership shown.	Combined refunds displayed correctly.	Passed	
Search Payment: Search transactions by receipt number.	Matching transactions displayed.	Correct results shown.	Passed	
Search Payment: Search transactions by customer name.	Matching transactions displayed.	Correct results shown.	Passed	
Filter Payment: Filter by payment method.	Only transactions with selected payment method displayed.	Filtered correctly.	Passed	

Group Name:	AbsTrack Fitness Gym: Gym Management System
Proponents:	Rosalada Justin Troy Merka Nathaniel Keene Felecilda Lloyd Justin Sexcion Eulo Icon
Module Name:	Inventory Management System

CL17L: SOFTWARE ENGINEERING 2

Milestone 6: Functional test case specification

SUMMARY OF TEST CASES EXECUTED

SCENARIOS	EXPECTED RESULT	ACTUAL RESULT	TEST RESULT (Passed/Failed)	ACTION TAKEN
Add Product: Can't add product if product number is empty.	Display validation error: "Product number is required."	Validation error displayed correctly.	Passed	
Add Product: Can't add product if product number already exists.	Display validation error: "This product number already exists."	Duplicate product number error displayed.	Passed	
Add Product: Can't add product if product name is empty.	Display validation error: "Product name is required."	Validation error displayed correctly.	Passed	
Add Product: Can't add product if category is empty.	Display validation error: "Category is required."	Validation error displayed correctly.	Passed	
Add Product: Can't add product if unit price is empty.	Display validation error: "Unit price is required."	Validation error displayed correctly.	Passed	
Add Product: Can't add product if unit price is not a number.	Display validation error: "Unit price must be a number."	Validation error displayed correctly.	Passed	

Add Product: Can't add product if unit price is negative.	Display validation error: "Unit price must be at least 0."	Validation error displayed correctly.	Passed	
Add Product: Can't add product if stock quantity is empty.	Display validation error: "Stock quantity is required."	Validation error displayed correctly.	Passed	
Add Product: Can't add product if stock quantity is not a whole number.	Display validation error: "Stock quantity must be a whole number."	Validation error displayed correctly.	Passed	
Add Product: Can't add product if stock quantity is negative.	Display validation error: "Stock quantity must be at least 0."	Validation error displayed correctly.	Passed	
Add Product: Can't add product if low stock threshold is empty.	Display validation error: "Low stock threshold is required."	Validation error displayed correctly.	Passed	
Add Product: Add product successfully with stock > 0 (sets last_restocked and creates initial transaction).	Display: "Product added successfully!" Product added with last_restocked timestamp and initial stock-in transaction.	Product added successfully with transaction log.	Passed	
Add Product: Add product successfully with stock = 0 (no last_restocked, no initial transaction).	Display: "Product added successfully!" Product added without last_restocked or transaction.	Product added successfully without transaction.	Passed	
Stock In: Stock in successfully increases stock quantity.	Display: "Stock added successfully!" Stock quantity increased.	Stock increased and success message shown.	Passed	
Stock In: Last restocked date updated after stock in.	last_restocked timestamp updates to current time.	Timestamp updated correctly.	Passed	
Stock Out: Stock out successfully decreases stock	Display: "Stock removed successfully!" Stock quantity	Stock decreased and success message shown.	Passed	

quantity.	decreased.			
Stock Out: Can't stock out if quantity exceeds available stock.	Display error: "Insufficient stock! Current stock: [n]."	Insufficient stock error displayed.	Passed	
Stock Out: Last restocked date NOT updated after stock out.	last_restocked remains unchanged.	Timestamp not changed.	Passed	
Stock Transaction: Can't process if quantity is 0.	Display validation error: quantity must be at least 1.	Validation error displayed.	Passed	
Stock Transaction: Can't process if quantity is negative.	Display validation error: quantity must be at least 1.	Validation error displayed.	Passed	
Delete Product: Delete product successfully if confirmed.	Display: "Product deleted successfully!"	Product deleted successfully.	Passed	
Delete Product: Can't delete product that doesn't exist.	Display error: Product not found.	Error displayed correctly.	Passed	
Bulk Delete: Can't bulk delete if no products are selected.	Display validation error for empty selection.	Warning displayed correctly.	Passed	
Bulk Delete: Bulk delete selected products if confirmed.	Display: "[n] product(s) deleted successfully!"	Products deleted and count shown.	Passed	
View Transaction History: View transaction history for a product.	Transaction history page displays all stock in/out records in descending order.	History displayed correctly.	Passed	
Search Inventory: Search products by product number.	Matching products displayed.	Correct results shown.	Passed	

Search Inventory: Search products by product name.	Matching products displayed.	Correct results shown.	Passed	
Search Inventory: Search products by category.	Matching products displayed.	Correct results shown.	Passed	
Filter Inventory: Sort products by name ascending.	Products sorted A-Z by name.	Sorted correctly.	Passed	
Filter Inventory: Sort products by stock quantity ascending.	Products sorted by lowest stock first.	Sorted correctly.	Passed	
KPI Dashboard: Total products count displayed correctly.	Displays correct total number of products.	Correct count displayed.	Passed	
KPI Dashboard: Low stock items count displayed correctly.	Displays count of items where stock < threshold and stock > 0.	Correct count displayed.	Passed	
KPI Dashboard: Out of stock items count displayed correctly.	Displays count of items where stock = 0.	Correct count displayed.	Passed	

Group Name:	AbsTrack Fitness Gym: Gym Management System
Proponents:	Rosalada Justin Troy Merka Nathaniel Keene Felecilda Lloyd Justin Sexcion Eulo Icon
Module Name:	Reports and Analytics Management System

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SUMMARY OF TEST CASES EXECUTED

SCENARIOS	EXPECTED RESULT	ACTUAL RESULT	TEST RESULT (Passed/Failed)	ACTION TAKEN
View Reports: Load reports page successfully.	Reports page displayed with all charts and KPI cards.	Page loaded correctly.	Passed	
KPI Cards: Monthly revenue displays correct total (retail + membership + PT).	KPI card shows accurate combined revenue for current month.	Correct revenue displayed.	Passed	
KPI Cards: Retail sales KPI displays correct amount.	Retail sales card shows sum of payment amounts for current month.	Correct amount displayed.	Passed	
KPI Cards: Membership revenue KPI displays correct amount.	Membership card shows sum of membership payments for current month.	Correct amount displayed.	Passed	
KPI Cards: PT revenue KPI displays correct amount (sessions × ₱500).	PT revenue card shows count of "done" sessions × 500.	Correct amount displayed.	Passed	
KPI Cards: Percentage change shows correct comparison vs previous	Percentage change calculated correctly (increase/decrease).	Correct percentage shown.	Passed	

month.				
KPI Cards: Percentage change shows 100% when previous month is 0 and current > 0.	Displays "100%" increase.	100% displayed correctly.	Passed	
KPI Cards: Percentage change shows 0% when both months are 0.	Displays "0%".	0% displayed correctly.	Passed	
Revenue Over Time Chart: Select "This Year" period.	Line chart shows monthly revenue breakdown for the year.	Chart displayed correctly.	Passed	
Revenue Over Time Chart: Select "Last 3 Months" period.	Line chart shows last 3 months of data.	Chart displayed correctly.	Passed	
Revenue Over Time Chart: Select "This Month" period.	Line chart shows current month data.	Chart displayed correctly.	Passed	
Top Selling Products Chart: Displays top 4 products by quantity sold.	Bar chart shows top 4 products with daily breakdown.	Chart displayed correctly.	Passed	
Top Selling Products Chart: No sales data available for selected period.	Chart displays empty/zero data gracefully.	Empty chart shown correctly.	Passed	
Revenue Breakdown Chart: Donut chart shows distribution across Retail, Membership, PT, and Walk-ins.	Donut chart displays 4 segments with correct amounts.	Chart displayed correctly.	Passed	
Transaction History Chart: Pie chart shows transactions grouped by payment method	Pie chart shows segments for each payment method used.	Chart displayed correctly.	Passed	

(Cash, Gcash, etc.).				
Customer Attendance Chart: Select "Today" period.	Line chart shows hourly attendance from 6 AM to 10 PM.	Chart displayed correctly.	Passed	
Customer Attendance Chart: Select "This Week" period.	Line chart shows daily attendance for the current week.	Chart displayed correctly.	Passed	
Export Report: Export report as PDF successfully.	PDF file downloads with report data.	PDF downloaded correctly.	Passed	
Export Report: Export report as CSV successfully.	CSV file downloads with report data.	CSV downloaded correctly.	Passed	
Export Report: Export report as Excel successfully.	Excel (XLS) file downloads with report data.	Excel downloaded correctly.	Passed	
Export Report: Can't export with unsupported format (e.g., "png").	Display error: "Unsupported export format."	Error displayed correctly.	Passed	
Export Report: Export with scope "all" includes KPIs, revenue, products, breakdown, transactions, attendance.	All sections included in exported file.	All data present in export.	Passed	
Export Report: Export with scope "revenue" includes only revenue data and KPIs.	Only revenue section and KPIs in exported file.	Correct data exported.	Passed	
Date Range Filter: "Today" returns data for current day only.	Reports show only today's data.	Correct date range applied.	Passed	

Date Range Filter: "This Week" returns data for current week.	Reports show current week's data.	Correct date range applied.	Passed	
Date Range Filter: "This Month" returns data for current month.	Reports show current month's data.	Correct date range applied.	Passed	
Date Range Filter: "Last Month" returns data for previous month.	Reports show previous month's data.	Correct date range applied.	Passed	
Date Range Filter: Invalid/unrecognized period defaults to "This Month".	Reports default to current month data.	Defaulted correctly.	Passed	

White Box Testing

Membership Management Test Cases

Test Case ID	Feature/Module	Testing Technique	Preconditions	Test Input	Expected Output	Actual Output	Status
WBT_MEM_001	calculate Status	Branch Coverage	Today is 2026-02-15	start_date: 2026-01-01, due_date: 2026-02-10 (daysUntilDue = -5)	'Expired'	'Expired'	Pass
WBT_MEM_002	calculate Status	Branch Coverage	Today is 2026-02-15	start_date: 2026-01-01, due_date: 2026-02-14 (boundary: daysUntilDue = -1)	'Expired'	'Expired'	Pass
WBT_MEM_003	calculate Status	Branch Coverage	Today is 2026-02-15	start_date: 2026-01-01, due_date: 2026-02-15 (boundary: daysUntilDue = 0, ≤ 7)	'Due soon'	'Due soon'	Pass
WBT_MEM_004	calculate Status	Branch Coverage	Today is 2026-02-15	start_date: 2026-01-01, due_date: 2026-02-16 (daysUntilDue = 1, ≤ 7)	'Due soon'	'Due soon'	Pass

WBT_ MEM_ 005	calculate Status	Branch Coverage	Today is 2026-02-15	start_date: 2026-01-01, due_date: 2026-02-22 (boundary: daysUntilDue = 7, ≤ 7)	'Due soon'	'Due soon'	Pass
WBT_ MEM_ 006	calculate Status	Branch Coverage	Today is 2026-02-15	start_date: 2026-01-01, due_date: 2026-02-23 (daysUntilDue = 8, > 7)	'Active'	'Active'	Pass
WBT_ MEM_ 007	calculate Status	Branch Coverage	Today is 2026-02-15	start_date: 2026-01-01, due_date: 2026-06-15 (daysUntilDue = 120, far future)	'Active'	'Active'	Pass
WBT_ MEM_ 008	calculate Status	Branch Coverage	Today is 2026-02-15	start_date: null, due_date: "invalid-date" (triggers exception)	'Active' (default on exception)	'Active'	Pass
WBT_ MEM_ 009	checkSim ilarName s	Branch Coverage	"John Doe" already exists in DB	name: "John Doe" (exact duplicate)	['type' => 'exact']	['type' => 'exact']	Pass
WBT_ MEM_ 010	checkSim ilarName s	Branch Coverage	"John Doe" already exists in DB	name: "John Smith" (same first, different last)	['type' => 'similar']	['type' => 'similar']	Pass

Attendance Management Test Cases

Test Case ID	Feature/Module	Testing Technique	Preconditions	Test Input	Expected Output	Actual Output	Status
WBT_ATT_001	storeAttendance	Branch Coverage	Valid client (ID=1) exists, no attendance today	client_id: 1, date: 2026-02-15, time_in: 08:00	{ success: true, message: 'Attendance recorded successfully!' }	{ success: true }	Pass
WBT_ATT_002	storeAttendance	Branch Coverage	client_id missing	client_id: null, date: 2026-02-15, time_in: 08:00	422 validation error: 'Please select a client.'	422 error	Pass
WBT_ATT_003	storeAttendance	Branch Coverage	client_id does not exist in clients table	client_id: 9999, date: 2026-02-15, time_in: 08:00	422 validation error: 'Selected client does not exist.'	422 error	Pass
WBT_ATT_004	storeAttendance	Branch Coverage	Client (ID=1) already checked in today	client_id: 1, date: 2026-02-15, time_in: 09:00	422: 'This client has already checked in today.'	422 duplicate	Pass
WBT_ATT_005	storeAttendance	Branch Coverage	Client status = 'Active'	client_id: 1 (status='Active'), date, time_in	status stored as 'active'	'active'	Pass
WBT_ATT_006	storeAttendance	Branch Coverage	Client status = 'Due soon'	client_id: 2 (status='Due soon'), date, time_in	status stored as 'due_soon' (converted)	'due_soon'	Pass
WBT_ATT_	storeAttendance	Branch Coverage	Client status = 'Expired'	client_id: 3 (status='Expired'), date,	status stored as 'expired'	'expired'	Pass

007				time_in			
WBT_ATT_008	storeAttendance	Branch Coverage	Client status = null	client_id: 4 (status=null), date, time_in	status defaults to 'active' (via ??)	'active'	Pass
WBT_ATT_009	getKPIs (% Change)	Branch Coverage	Yesterday attendance = 10, Today = 15	customersYesterday: 10, customersToday: 15	50.0 (50% increase)	50.0	Pass
WBT_ATT_010	getKPIs (% Change)	Branch Coverage	Yesterday attendance = 0	customersYesterday: 0, customersToday: 5	0 (division by zero guard)	0	Pass
WBT_ATT_011	getKPIs (% Change)	Branch Coverage	Yesterday = 20, Today = 10	customersYesterday: 20, customersToday: 10	-50.0 (50% decrease)	-50.0	Pass
WBT_ATT_012	getKPIs (% Change)	Branch Coverage	Yesterday = 0, Today = 0	customersYesterday: 0, customersToday: 0	0	0	Pass

Payment & Billing Management Test Cases

Test Case ID	Feature/Module	Testing Technique	Preconditions	Test Input	Expected Output	Actual Output	Status
WBT_PAY_001	store (Process Payment)	Branch Coverage	Valid items in stock, sufficient payment	customer: "John", paid: 500.00, total: 450.00, items: [{id:1, qty:2, price:225}]	Success: return_amount = 50.00, stock decremented	Success, stock updated	Pass
WBT_PAY_002	store (Process Payment)	Branch Coverage	paid_amount < total_amount	customer: "John", paid: 100.00, total: 450.00	Error: 'Paid amount must be equal to or greater than total amount.'	Validation error	Pass
WBT_PAY_003	store (Process Payment)	Branch Coverage	paid_amount = total_amount (exact)	customer: "John", paid: 450.00, total: 450.00, items valid	Success: return_amount = 0.00	0.00 return	Pass
WBT_PAY_004	store (Process Payment)	Branch Coverage	Inventory item stock < requested qty	item stock_qty: 2, requested qty: 5	400: 'Insufficient stock for [product]'	400 abort	Pass
WBT_PAY_005	store (Process Payment)	Branch Coverage	Inventory item stock = requested qty (boundary)	item stock_qty: 5, requested qty: 5	Success: new stock = 0	stock = 0	Pass
WBT_PAY_	store (Process	Branch	customer_nam	customer: null	422 Validation error:	422 error	Pass

006	Payment)	Coverage	e field missing		'required'		
WBT_ PAY_ 007	store (Process Payment)	Branch Coverage	items_data is empty JSON	items_data: "[]"	Validation passes but no items processed	Empty payment	Pass
WBT_ PAY_ 008	store (Process Payment)	Branch Coverage	AJAX request (expectsJson = true)	Valid data, sent via AJAX	JSON: { success: true, message: 'Payment completed successfully.' }	JSON response	Pass
WBT_ PAY_ 009	store (Process Payment)	Branch Coverage	Standard form submission (non-AJAX)	Valid data, form POST	Redirect to payments.index with success flash	Redirect	Pass
WBT_ PAY_ 010	store (Process Payment)	Branch Coverage	Overpayment scenario	paid: 1234.56, total: 1000.00	return_amount = 234.56	234.56	Pass
WBT_ PAY_ 011	destroy (Delete Payment)	Branch Coverage	Payment ID=1 exists with 2 items, inventory exists	payment: 1	Success: stock restored, payment + items deleted	Success	Pass
WBT_ PAY_ 012	destroy (Delete Payment)	Branch Coverage	Payment ID=999 does not exist	payment: 999	Exception: 'Failed to delete transaction: ...'	Error caught	Pass

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Inventory Management System Test Cases

Test Case ID	Feature/Module	Testing Technique	Preconditions	Test Input	Expected Output	Actual Output	Status
WBT_INV_001	stockTransaction (Stock In)	Branch Coverage	Item ID=1, current stock = 10	type: 'stock_in', quantity: 5	newStock = 15, last_restocked updated	15, restocked updated	Pass
WBT_INV_002	stockTransaction (Stock Out - sufficient)	Branch Coverage	Item ID=1, current stock = 10	type: 'stock_out', quantity: 3	newStock = 7, last_restocked NOT updated	7, no restock update	Pass
WBT_INV_003	stockTransaction (Stock Out - exact)	Branch Coverage	Item ID=1, current stock = 10	type: 'stock_out', quantity: 10 (boundary)	newStock = 0	0	Pass
WBT_INV_004	stockTransaction (Stock Out - insufficient)	Branch Coverage	Item ID=1, current stock = 5	type: 'stock_out', quantity: 10	Error: 'Insufficient stock! Current stock: 5'	Error returned	Pass

WBT_INV_005	stockTransaction (Stock In - min valid)	Branch Coverage	Item ID=1, current stock = 0	type: 'stock_in', quantity: 1 (minimum valid)	newStock = 1, last_restocked set	1	Pass
WBT_INV_006	stockTransaction (Zero quantity)	Branch Coverage	N/A	type: 'stock_in', quantity: 0	422 Validation error: 'min:1'	Validation error	Pass
WBT_INV_007	stockTransaction (Negative quantity)	Branch Coverage	N/A	type: 'stock_in', quantity: -5	422 Validation error: 'min:1'	Validation error	Pass
WBT_INV_008	stockTransaction (DB failure)	Branch Coverage	DB transaction throws exception	type: 'stock_in', quantity: 5 (simulate DB fail)	Error: 'Transaction failed: ...', DB rolled back	Rollback + error	Pass
WBT_INV_009	stockTransaction (Large stock in)	Branch Coverage	Item ID=1, current stock = 100	type: 'stock_in', quantity: 9999	newStock = 10099	10099	Pass
WBT_INV_010	store (Add Product - stock > 0)	Branch Coverage	Valid data, stock_qty > 0	product_number: 'PRD-100', stock_qty: 25	Product created, last_restocked set, initial transaction created	Success + transaction	Pass

WBT_INV_011	store (Add Product - stock = 0)	Branch Coverage	Valid data, stock_qty = 0	product_number: 'PRD-101', stock_qty: 0	Product created, NO last_restocked, NO initial transaction	Success, no transaction	Pass
WBT_INV_012	store (Duplicate product number)	Branch Coverage	product_number 'PROD001' already exists	product_number: 'PROD001'	Validation error: 'This product number already exists.'	422 unique error	Pass
WBT_INV_013	store (Negative unit_price)	Branch Coverage	N/A	unit_price: -10	Validation error: 'min:0'	Validation error	Pass

Reports & Analytics Management Test Cases

Test Case ID	Feature/Module	Testing Technique	Preconditions	Test Input	Expected Output	Actual Output	Status
WBT_RPT_001	calculate PercentChange	Branch Coverage	N/A	oldValue: 0, newValue: 500 (old=0, new>0)	100	100	Pass
WBT_RPT_002	calculate PercentChange	Branch Coverage	N/A	oldValue: 0, newValue: 0 (old=0, new=0)	0	0	Pass
WBT_RPT_003	calculate PercentChange	Branch Coverage	N/A	oldValue: 1000, newValue: 1500 (50% increase)	50.0	50.0	Pass
WBT_RPT_004	calculate PercentChange	Branch Coverage	N/A	oldValue: 1000, newValue: 500 (50% decrease)	-50.0	-50.0	Pass
WBT_RPT_005	calculate PercentChange	Branch Coverage	N/A	oldValue: 1000, newValue: 1000 (no change)	0.0	0.0	Pass
WBT_RPT_006	calculate PercentChange	Branch Coverage	N/A	oldValue: 200, newValue: 0 (100% decrease)	-100.0	-100.0	Pass
WBT_RPT_007	calculate PercentChange	Branch Coverage	N/A	oldValue: 100, newValue: 333	233.0	233.0	Pass

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WBT_RPT_008	getDateRange	Branch Coverage	Today is 2026-02-15	period: 'today'	2026-02-15 00:00:00 → 2026-02-15 23:59:59	Correct range	Pass
WBT_RPT_009	getDateRange	Branch Coverage	Today is 2026-02-15	period: 'this_week'	Start of week → End of week	Correct range	Pass
WBT_RPT_010	getDateRange	Branch Coverage	Today is 2026-02-15	period: 'this_month'	2026-02-01 → 2026-02-28	Correct range	Pass
WBT_RPT_011	getDateRange	Branch Coverage	Today is 2026-02-15	period: 'last_month'	2026-01-01 → 2026-01-31	Correct range	Pass
WBT_RPT_012	getDateRange	Branch Coverage	Today is 2026-02-15	period: 'last_3_months'	2025-12-01 → 2026-02-28	Correct range	Pass
WBT_RPT_013	getDateRange	Branch Coverage	Today is 2026-02-15	period: 'this_year'	2026-01-01 → 2026-12-31	Correct range	Pass
WBT_RPT_014	exportReport	Branch Coverage	Valid data	format: 'pdf', scope: 'all'	PDF file downloaded	PDF response	Pass
WBT_RPT_015	exportReport	Branch Coverage	Valid data	format: 'csv', scope: 'kpis'	CSV file downloaded	CSV response	Pass

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WBT_RPT_016	exportReport	Branch Coverage	Valid data	format: 'excel', scope: 'revenue'	Excel (XLS) file downloaded	XLS response	Pass
WBT_RPT_017	exportReport	Branch Coverage	N/A	format: 'png' (unsupported, hits default)	400: 'Unsupported export format'	400 error	Pass
WBT_RPT_018	exportReport	Branch Coverage	Exception during gather	format: 'pdf' (simulate DB error)	500: 'Failed to export report: ...'	500 error	Pass